

ABU NAYEM MD. ASRAF SIDDIQUEE

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🌐 [Abu Nayem Md. Asraf Siddiquee](#)

📺 [Asraf's Garage](#)

Summary

Mechanical and robotics engineer with strong expertise in mechanical design, simulation, and experimental validation of robotic and electromechanical systems. Proficient in CAD-driven product development, nonlinear FEA, multiphysics simulation, and physics-informed modeling, with hands-on experience in embedded hardware prototyping, sensor integration, and additive manufacturing. Experienced in leading research-driven hardware projects across underwater robotics, soft robotics, and energy systems, from concept through testing and deployment.

Education

Ph.D. Candidate in Electrical Engineering [University of Notre Dame, Notre Dame, IN]	Expected Graduation: 2027
M.S. in Electrical Engineering [University of Notre Dame, Notre Dame, IN]	2022 - 2023
Master of Engineering Management [Arkansas State University, Jonesboro, AR]	2021 - 2022
M.S. in Mechanical Engineering [Arkansas State University, Jonesboro, AR]	2015 - 2017
B.S. in Mechanical Engineering [Khulna University of Engineering & Technology, Bangladesh]	2009 - 2013

Research Experience

miNiRo Lab, University of Notre Dame GRADUATE RESEARCH & TEACHING ASSISTANT

Notre Dame, IN, USA
Aug. 2022 – Present

- Led a 6-month **Department of Defense (DoD)-funded project**, developing an advanced underwater burrowing robot from conceptual design through fabrication and experimental validation, demonstrating adaptive locomotion in complex terrains.
- Designed and developed **bio-inspired soft robotic systems**, including multi-legged platforms for terrestrial and shallow-water exploration, enabling applications in environmental monitoring, search and rescue, disaster-stricken areas, defense, and infrastructure inspection.
- Engineered **multi-material soft robotic structures and actuators**, integrating elastomers, composites, and embedded components to enhance mobility, adaptability, and functional performance.
- Performed **shape optimization and finite element analysis (FEA)** to improve robotic performance; applied **physics-informed machine learning** for ongoing design optimization in complex environments.
- Mentored **13 graduate and visiting student researchers**, providing independent guidance on experimental design, fabrication, and simulation, resulting in 10+ conference presentations and research outputs.
- Applied multi-disciplinary engineering expertise to integrate **mechanical design, embedded control, and autonomous system development**, contributing to high-impact research outcomes.

Thermal Engineering Lab, Arkansas State University POSTGRADUATE RESEARCH ASSISTANT

Jonesboro, AR, USA
Dec. 2017 – Aug. 2018

- Led development of an **analytical model** to predict the performance of a wire-plate type **electrostatic precipitator (ESP)** in an **870 MW supercritical coal-fired power plant**, analyzing NOX removal efficiency of a **selective catalytic reduction (SCR) system**.
- Contributed to a **\$1,220,000 project** on predictive modeling for transient ash deposition in post-boiler heat exchangers, including data analysis, modeling, and validation.

Thermal Engineering Lab, Arkansas State University
GRADUATE RESEARCH ASSISTANT

Jonesboro, AR, USA
Jan. 2015 – Dec. 2017

- Project 1 (**ASTA Basic Research Award, \$79,379**): Developed an analytical model using **Maple 2015** to predict transient behavior of **vanadium redox flow batteries (VRFB)** for renewable power plant applications, evaluating the effect of operating conditions on VRFB performance.
- Project 2 (**NSF EPSCoR, \$120,000**): Co-managed synthesis, setup, and data collection for an **advanced flow battery test-bed** for renewable energy storage, contributing to design and operational evaluation.

Academic Experience

University of Notre Dame
GRADUATE TEACHING ASSISTANT

Notre Dame, IN, USA
Fall 2023 – Fall 2025

- Mentored student teams in **Experimental Robotics (EE 47072)**, supervising the design, fabrication, and experimental validation of five robotic systems including bio inspired swimmers, jumping frogs, inchworms, and tree climbing robots, while introducing computational modeling, simulation, and experimental design concepts for **soft robotics** and autonomous systems.
- **Intro to Electrical Engineering Lab (Fall 2023, Fall 2025)**: Supervised laboratory experiments and evaluated lab reports for two sections of 20 students each, supporting student learning in fundamental electrical engineering concepts and experimental methods.

Arkansas State University
GRADUATE TEACHING ASSISTANT

Jonesboro, AR, USA
Fall 2021 – Spring 2022

- ENGR 3473 – **Fluid Mechanics**: Graded homework and supported course instruction via student consultations and problem-solving sessions.
- ENGR 3471 – **Fluid Mechanics Laboratory**: Assisted in supervising lab experiments, guiding data analysis, and evaluating lab reports.

Arkansas State University
INSTRUCTOR OF MECHANICAL ENGINEERING

Jonesboro, AR, USA
Aug. 2018 – May 2019

- Delivered undergraduate instruction in core mechanical engineering courses with emphasis on applied problem-solving and hands-on analysis:
 - ENGR 1402**: Concepts of Engineering [Fall-2018]
 - ENGR 3423**: Dynamics [Fall-2018, Spring-2019]
 - ENGR 3443**: Engineering Thermodynamics I [Spring-2019]
 - ENGR 3471**: Fluid Mechanics Laboratory [Fall-2018]
- Supervised and evaluated **Arduino-based mechatronic design projects** for 17 undergraduate students and served as a technical judge at the A-State Arduino Fair.
- Contributed to **ABET accreditation activities** by aligning course syllabi, assessment rubrics, and learning outcomes with program-level objectives.
- Integrated **SolidWorks simulation modules** into the Thermodynamics I curriculum, introducing students to computational heat transfer concepts and applied engineering simulation tools.

Industry Experience

AceOne Technologies HARDWARE ENGINEER INTERN

Jonesboro, AR, USA
Jan. 2020 – Jul. 2021

- Developed embedded **C++ and Arduino firmware** for real-time acquisition and control of servo motors, vibration, IR, current, temperature, and pressure sensors, with cloud-based data logging.
- Collaborated with the product engineering team to design and validate **PackRight smart device hardware**, including enclosure redesign and integration of high-accuracy temperature sensors.
- Implemented **GPS and Bluetooth communication modules** for multi-sensor data fusion and remote monitoring.
- Designed, prototyped, and validated **3D-printed electromechanical components**, including wall-mount brackets for electrical installations.
- Redesigned and fabricated **Montana Mask prototypes** using additive manufacturing to improve ergonomics and usability during the COVID-19 pandemic.

A-State Innovate, The Garage MECHANICAL ENGINEER (VOLUNTEER)

Jonesboro, AR, USA
May 2019 – Jan. 2020

- Designed and fabricated **soft robotic structures, mechanical assemblies, and prototype components** using fused deposition modeling (FDM) and epoxy resin casting.
- Applied **finite element analysis (FEA)**, geometric dimensioning and tolerancing (**GD&T, ASME Y14.5**), and design for additive manufacturing (**DFAM**) principles to improve structural integrity and manufacturability.
- Documented end-to-end design workflows supporting robotics and product development, including **CAD models, soft robotics prototypes, and additive manufacturing builds**.

Adroit Engineers MECHANICAL DESIGN & PROTOTYPING ENGINEER

Dhaka, Bangladesh
Oct. 2013 – Jan. 2015

- Led mechanical design and **FEA of electromechanical medical devices**, including an automated wheelchair and an electric hospital bed.
- Designed printed circuit boards (**PCBs**) and developed embedded **C/C++ control software** for biomedical devices such as an RPM calculator and temperature sensing module.
- Optimized automated wheelchair structural design, reducing mass and manufacturing cost by 7% while maintaining safety factors.
- Managed **multi-disciplinary teams**, bills of materials (BOM), and design validation workflows as project manager for medical and consumer electronic devices.

Publications

JOURNAL ARTICLES

- **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. Design and Simulation of Soft, Multi-Legged Robotic Swimmer for Shallow Water Locomotion. [Under Review].
- **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. (2024). Utilizing Bioinspired Soft Modular Appendages for Grasping and Locomotion in Multi-Legged Robots on Ground and Underwater. IEEE Robotics and Automation Letters (RA-L).

CONFERENCE PAPERS

- **Siddiquee, A. N. M. A.**, Castillo Gonzalez, M., Utama, V., Zhu, Y., & Ozkan-Aydin, Y. (2026). A Modular, Sea Star-Inspired Robot with Autonomous Rays: Autotomy-driven Robustness through Morphological Design. [Under Review]
- **Siddiquee, A.**, & Jeong, K. (2016). Conjugated Dynamic Modeling on Vanadium Redox Flow Battery With Non-Constant Variance for Renewable Power Plant Applications. Proceedings of the ASME 2016 International Mechanical Engineering Congress and Exposition (IMECE2016-67462), Phoenix, AZ, USA.
- Jeong, K., Saifujjaman, M., Aryal, S., Tamang, S., **Siddiquee, A.**, & Lee, S. (2019). Modeling for Blended Coal Combustion and its Impact on Ash Deposition in Full-Scale Post-Boiler Equipment in a Supercritical Pulverized Coal-fired Power Plant. Proceedings of the 12th Asia-Pacific Conference on Combustion.

ONLINE RESEARCH PROFILES

- Google Scholar: Abu Nayem Md. Asraf Siddiquee.
- YouTube Channel: Asraf's Garage – robotics projects and demonstrations.

Oral Presentations

7. **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. (2026, January). Design and simulation of soft, multi-legged robotic swimmer for shallow water locomotion. *Society for Integrative and Comparative Biology (SICB) 2026 Annual Meeting*, Portland, OR, USA.
6. Castillo Gonzalez, M., **Siddiquee, A. N. M. A.**, Zhu, Y., Ertas, M. S., & Ozkan-Aydin, Y. (2026, January). The impact of passive anisotropic friction on the locomotion of modular, sea star-inspired robot. *SICB 2026 Annual Meeting*, Portland, OR, USA.
5. **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. (2025, May). Utilizing bioinspired soft modular appendages for grasping and locomotion in multi-legged robots on ground and underwater. *IEEE International Conference on Robotics and Automation (ICRA)*, Atlanta, GA, USA.
4. **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. (2025, January). Bioinspired, modular, hybrid quadruped robot with multifunctional appendages for enhanced underwater exploration and object manipulation. *SICB 2025 Annual Meeting*, Atlanta, GA, USA.
3. **Siddiquee, A.**, Jeong, K., Lee, E., & Lee, S. (2018, November). Performance prediction modeling of a full-scale electrostatic precipitator (IMECE2018-89748). *ASME International Mechanical Engineering Congress & Exposition*, Pittsburgh, PA, USA.
2. **Siddiquee, A.**, Jeong, K., & Lee, J. (2017, November). Modeling for performance prediction of electrostatic precipitator (IMECE2017-73248). *ASME IMECE*, Tampa, FL, USA.
1. **Siddiquee, A.**, & Jeong, K. (2016, November). Conjugated dynamic modeling on vanadium redox flow battery with non-constant variance for renewable power plant applications (IMECE2016-67462). *ASME IMECE*, Phoenix, AZ, USA.

Poster Presentation

8. **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. (2026, June). Design, simulation, and locomotion analysis of a modular sea star-inspired robotic ray with passive anisotropic friction. *Midwest Robotics Workshop (MWRW)*, Toyota Technological Institute at Chicago, Chicago, IL, USA.
7. **Siddiquee, A. N. M. A.**, Castillo Gonzalez, M., & Ozkan-Aydin, Y. (2026, January). Simulation and design of the sea star-inspired robot ray. *SICB 2026 Annual Meeting*, Portland, OR, USA.
6. **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. (2025, June). Design and simulation of soft, multi-legged robotic swimmer for shallow water locomotion. *Midwest Robotics Workshop (MWRW)*, Toyota Technological Institute at Chicago, Chicago, IL, USA.
5. **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. (2025, March). Cross-Environmental Locomotion: Bioinspired Soft Modular Robot for Advanced Search and Rescue Operations. *6th Annual Indiana Science Communication Day, Indiana Statehouse*, Indianapolis, IN, USA. (Invited research presentation at a statewide science policy and innovation forum.)
4. **Siddiquee, A. N. M. A.** (2024). Bioinspired, modular, hybrid quadruped robot with multifunctional appendages for enhanced underwater exploration and object manipulation. *ICON Student Research Conference*, Purdue University, West Lafayette, IN, USA.
3. **Siddiquee, A. N. M. A.**, & Ozkan-Aydin, Y. (2024, April). Design and development of bioinspired soft modular appendages for object manipulation and locomotion in multi-legged robots across terrestrial and aquatic environments. *Midwest Robotics Workshop (MWRW)*, Toyota Technological Institute at Chicago, Chicago, IL, USA.
2. **Siddiquee, A.**, & Jeong, K. (2015, September). Conjugated dynamic modeling with non-constant variance on vanadium redox flow battery for renewable power plant. *NSF EPSCoR Annual Meeting*, Fayetteville, AR, USA.

1. Islam, R., **Siddiquee, A.**, Nolen, C., Jeong, K., & McCann, R. (2015, September). Test-bed development for vanadium redox flow battery (VRFB). *NSF EPSCoR Annual Meeting*, Fayetteville, AR, USA.

Graduate Mentorship

Mentored visiting scholars, undergraduate researchers, and laboratory interns, guiding original research from conceptual design through fabrication, experimentation, and dissemination. Mentorship focused on the design and development of **bio-inspired robots, soft actuators, and electro-mechanical systems**, resulting in **10+ oral and poster presentations**. Mentorship was provided under the supervision of Dr. Yasemin Ozkan-Aydin (University of Notre Dame), with **independent day-to-day technical, experimental, and research guidance**.

Mentees

13. Raafe Rahman, High School Research Intern (Summer 2024–Summer 2025, Summer 2026), **Currently pursuing a B.S. in ME at Purdue University**
12. Duolan Sheng, University at Stony Brook (Summer 2025)
11. Aria Meyer, Electrical Engineering, University of Notre Dame (Summer 2025)
10. Robert John Urquhart, University of Galway (Summer 2025)
9. Ishmael Auqui, Electrical Engineering, University of Notre Dame (Spring & Fall 2025)
8. Minjae Kim, Electrical Engineering, University of Notre Dame (Spring 2025)
7. Yuhan Zhu, Visiting Student (China), University of Notre Dame (Spring 2025)
6. Cole Descalzi, Electrical Engineering, University of Notre Dame (Spring 2025)
5. Vanessa Utama, Electrical Engineering, University of Notre Dame (Spring 2025)
4. Martín Eduardo Castillo González, Pontificia Universidad Católica de Chile (Jan–Mar 2025)
3. Kyle McKee, Electrical Engineering, University of Notre Dame (Spring 2024)
2. Nagham Bou Hamdan, Electrical Engineering, Notre Dame University–Louaize (Summer 2024)
1. Ben Colfer, Electrical Engineering, University of Notre Dame (Spring 2023)

Professional Service

PEER REVIEW AND CONFERENCE SERVICE

- Reviewer for **IEEE International Conference on Intelligent Robots and Systems (IEEE IROS, 2024)**.
- Volunteer at **IEEE IROS 2023** and **Society for Integrative and Comparative Biology (SICB) Annual Meeting, 2025**, assisting with sessions, demonstrations, and logistics.
- Coach at **Innovate-o-Thon**, University of Notre Dame (7 sessions), mentoring student teams by solving engineering and technical problems provided by **Marmon Group**, an American industrial holding company.
- Served as a **Judge** for the **2026 Northern Indiana Regional Science & Engineering Fair**, University of Notre Dame (February 28, 2026), evaluating research projects across the **Elementary (Grades 3–5)**, **Junior (Grades 6–8)**, and **Senior (Grades 9–12)** divisions under International Science and Engineering Fair (ISEF) judging criteria.
- Served as a **Judge** for the Engineering Undergraduate Poster Competition, **College of Science and Engineering Joint Annual Meeting (COSE-JAM)**, University of Notre Dame (2025, 2026).
- Served as a **technical judge** at the **A-State Arduino Fair**, Arkansas State University, December 2018.

PROFESSIONAL MEMBERSHIPS

- Student Member, **Institute of Electrical and Electronics Engineers (IEEE)**
- Student Member, **American Society of Mechanical Engineers (ASME)**
- Student Member, **Society for Integrative and Comparative Biology (SICB)**

Leadership & Community Service

LEADERSHIP ROLES

- Student Leader for the **Robotics and Controls Seminar Series**, Fall 2024, University of Notre Dame (Robotics @ND); coordinated seminars involving **15 faculty** and **30+ graduate students**.
- Founding Member and **Treasurer** of the **Bangladeshi Student Association**, Arkansas State University (2015), responsible for establishing the organization's financial structure and overseeing budget management.
- Organized orientation programs for new Bangladeshi students at Arkansas State University.
- President of **Greater Comilla Association**, Khulna University of Engineering & Technology (KUET), Bangladesh; led 120+ students, organizing tours, cultural events, and campfires (1-year term).
- Secretary of **Greater Comilla Association**, KUET (1-year term), coordinating events and student engagement.
- Class Representative, Khulna University of Engineering & Technology, 1 semester.

COMMUNITY OUTREACH & VOLUNTEER ACTIVITIES

- While serving as an **Instructor of Mechanical Engineering** at Arkansas State University, participated in a faculty-led fundraising effort and collected contributions from students, helping raise **\$220** to support medical expenses for a staff member's child undergoing cancer treatment.

Honors & Awards

GRADUATE AWARDS

- **Edison Innovation Fellowship**, University of Notre Dame IDEA Center (2026–2027) — Selected as one of 10 graduate fellows university-wide and one of five from the College of Engineering; awarded a Spring 2027 stipend of **\$17,077.50** to advance research commercialization, customer discovery, and technology derisking.
- Recipient of **TEEWS Graduate Incubator & Fellowship**, University of Notre Dame, recognizing research excellence and innovation in engineering.
- Travel Grant: **IEEE ICRA Conference Travel Grant 2025 (\$1,500)**.
- Travel Grant: **James L. Massey Travel Grant (3 × \$750)**, University of Notre Dame.
- **Innovate-o-Thon Judge Grant**, University of Notre Dame / Marmon Holdings Inc. (**7 × \$250**).
- Travel Grant: **Arkansas State University Travel Grant (\$1,500)**.
- Travel Grant: **Bangladesh Sweden Trust Fund-2016 (\$500)**.

UNDERGRADUATE & HIGH SCHOOL AWARDS

- **Technical Scholarship**, Bangladesh (2009–2013) — Complimentary government scholarship for regular engineering students.
- Recipient of **Math Olympiad of Bangladesh**, Regional 2nd Runner-Up (Top 72 nationwide, 2005).

LANGUAGE PROFICIENCY

- Fluent in **Bengali** and **English**; proficient in **Hindi** and **Urdu** (speaking and understanding).

Technical Skills

- **Mechanical Design & Fabrication:** SolidWorks, Fusion 360 (Parametric & Generative), Siemens NX, AutoCAD, Rhinoceros 3D; Rapid Prototyping (3D Printing, Laser Cutting); Multi-Material Integration.
- **Simulation & Analysis:** Abaqus, COMSOL, ANSYS (Fluent, Workbench); Nonlinear FEA; Multiphysics Modeling.
- **Programming:** Python, MATLAB, Maple.
- **Electronics & Embedded Systems:** Arduino, Embedded C++, Sensor Integration, Hardware Prototyping.